

Assignment 1 – Dataset Analysis & Data Cleaning Notes

This document summarizes all the important concepts, pandas commands, matplotlib basics, and analytical thinking patterns learned while solving Assignment 1 using the Titanic dataset.

1. Core Concepts Learned

- Dataset inspection and understanding
- Rows, columns, and DataFrame structure
- Null value detection and handling
- Difference between mean and median
- Duplicate row detection
- Feature selection and noisy columns
- Numerical vs categorical columns
- Grouped analysis using groupby
- Boolean logic in pandas (True = 1, False = 0)
- Visualization fundamentals using matplotlib
- Importance of interpreting outputs instead of memorizing syntax
- Working on copied dataframe instead of modifying original dataset

2. Important Pandas Functions and Use Cases

Command / Function	Purpose / Use Case
df.info()	Shows dataset structure, datatype, non-null values
df.describe()	Statistical summary of numerical columns
df.columns	Displays all column names
df.isnull()	Checks whether values are null or not
df.isnull().sum()	Counts total null values
df.isnull().any()	Checks whether any null value exists
df.copy()	Creates independent copy of dataframe
df.drop(columns, axis=1)	Removes columns from dataframe
df['Age'].median()	Finds median of Age column
fillna()	Fills missing values
duplicated()	Checks duplicate rows
value_counts()	Counts frequency of values
groupby()	Groups data by category
plt.bar()	Creates bar chart
plt.xlabel()	Sets X-axis label

plt.ylabel()	Sets Y-axis label
plt.title()	Adds chart title
plt.show()	Displays graph

3. Advanced Concepts Encountered During Assignment

- Understanding grouped output hierarchy
- Difference between count and ratio
- Importance of interpreting grouped data carefully
- Difference between reference and actual dataframe copy
- Why median is preferred for skewed numerical data
- Why graphs must represent logically consistent information
- Importance of verifying preprocessing steps
- Feature relevance in preprocessing
- Difference between raw data and cleaned data
- Data analysis as question-driven exploration

4. Key Mindset Learned

The most important learning from Assignment 1 is that data analysis is not about memorizing syntax. It is about understanding data, asking meaningful questions, interpreting outputs correctly, and making logical preprocessing decisions.